extensible platform for building next-generation processors.

Q What certifications and quality assurance testing has the chip gone through to ensure maximum usage?

A. SHAKTI has an open source-based test methodology and a comprehensive set of test suites. This is at par with standard industry practices. We are also working on formal verification techniques.

After the chip is fabricated, the foundry or entity that manufactures the chip does chip-level qualification testing. Then, we re-run our standards tests and some application tests to check for correctness, stability and performance.

Q Is the chip ready for such emerging technologies as 5G, machine learning (ML) or artificial intelligence (AI)?

A. IIT-M develops the basic cores and proves them on test chips. The startups then take these cores and build ASICs for the above-mentioned applications.

Q How competitive is the chip in terms of pricing?

A. In terms of development and IP cost, it is cheaper. Ultimately, cost of the chip is tied to its size. In that respect, SHAKTI cores are very optimal and competitive with contemporary cores.

Q How is open source playing a role in chip manufacturing?

A. IIT-M's computer science department is an advocate of open source and open standards for a long time, which led to the decision to co-found the RISC-V consortium. The open source chip movement has just begun; how it will change the world remains to be seen.

G.S. Madhusudan
senior project advisor, IIT-Madras

Q Who is RISC-V more beneficial to, chip designer or customer?

A. Both. For the designer, it allows easy design exploration, where an optimal design can be reached quicker.

For the customer, it offers a choice of vendors, since many vendors provide royalty-free RISC-V designs.

For example, a college kid with an idea can simply go to SHAKTI website, take Verilog code, compile it, test it out, make any required changes and within weeks create a new design.

Q What kind of application areas is RISC-V platform best suited for?

A. Pretty much any kind of design can be built with RISC-V since it is a general-purpose ISA.

Q Will students going to the SHAKTI website help in its growth?

A. Increased student use of SHAKTI will increase its adoption and ensure availability of RISC-V-trained engineers. This will help strengthen the CPU design ecosystem in India. We will soon launch low-cost FPGA kits to help students experiment with SHAKTI.

Q When will these kits come out?

A. The kits are ready, and should be available by September, once the software around them has been created.

Q Any startups that have adopted SHAKTI?

A. InCore Semiconductors uses some of the SHAKTI IP. The startup focuses on AI/ML systems, security and fault-tolerant computing.

Q Since the technology is open source, how do you monetise it?

A. Thales, one of the world's largest defence electronics company, is evaluating SHAKTI processors for aerospace applications, and has given a large grant to IIT-M to accelerate development. Many entities have started giving funds to IIT-M or are looking forward to collaborate to figure out how to use SHAKTI in their products.

Startups will provide commercial services on top of SHAKTI, or provide enhancements to SHAKTI. END

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